MILITARY SPECIFICATION

CHARGE, PROPELLING, 155MM, M3A1
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of propelling charge designated as Charge, Propelling, M3A1, for 155MM Cannon, M1, M45, M126, M185 and M199.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. Unless otherwise specified, the folowing specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1320

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection By Attributes

MIL-STD-414 - Sampling Procedures and Tables for Inspection By Variables For Percent Defective

MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures And Tables For Inspection By Attributes

2.1.2 Other government documents, drawings and publications. The following other Government documents form a part of this specification to the extent specified herein.

DRAWINGS

U.S ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

7548187 - Packing and Marking of Metal Containers, for Propelling Charges, Assembly and Details

8887277 - Charge, Propelling, 155MM, M3A1 for Cannon, M1, M45, M126, M185, M199

5539183 - Diameter and Length Gage

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

(copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

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3.2 Propelling charge. The propelling charge shall comply with all requirements specified on Drawing (dwg.) 8887277 and with all requirements specified in applicable specifications.

3.3 Weights

3.3.1 Propellant weight. The net weight of the propellant shall comply with the weights specified in the loading authorization within the following tolerances.

Increment Number	Tolerance Plus or Minus Ounces
1	0.05
2	0.02
3	0.02
4	0.05
5	0.05

3.3.2 Clean burning igniter (CBI) or ball powder igniter (BPI) weight. The net weight of CBI or BPI shall comply with the weight specified on the drawing.

3.3.3 Charge, propelling assembly weight. The weight of the propelling charge prior to packing shall not differ by more than ± .50 ounces from the total of the amount specified in the loading authorization plus actual or average weights of the empty increment and loaded base igniter and flash reducers.

3.4 Dimensional gage

3.4.1 Maximum length and diameter. The completed charge shall freely enter the prescribed gage. The gage shall be designed to inspect the maximum diameter and length.

3.5 Proving ground. The muzzle velocity, pressure and velocity standard deviation at 70°F shall be as specified in Table I.

Table I Velocity and Pressure

Zone	Velocity (fps) Minimum	Velocity (fps) Maximum	Velocity Std Dev (fps)	Individual Max Press (psi)
1	655	705	6.0	-
5	1205	1255	6.0	30,500

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3.6 Assembly torque. The packing container test plug shall be assembled to withstand subsequent reapplication of the minimum assembly torque specified on the applicable drawing, without moving.

3.7 First article inspection. This specification contains technical provisions for the first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.8 Workmanship. All components shall be free of dirt, grease and other foreign material. There shall be no discoloration of cloth and no evidence of rips, tears, or other cloth defects. All work shall be performed in a thorough workmanlike manner.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample from the initial production lot as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. This sample shall consist of the following:

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Part Description	Drawing	Quantity
Flash Reducer	8887288	5
Flash Reducer, Increment 1	8887286	5
Flash Reducer	8887292	5
Reducer, Flash	8887290	5
Igniter, End	8887283	5
Igniter Increment 1	8887281	5
Base Body Charge 1	9204942	5
Base Chg 1 Body & Igniter Incr Assembly	9204943/9295112	5
Base Charge 1 Loading Assembly	8887278/9295111	5
Body Increment	9221256	5
Increment Loading Assembly	9221257	5
Incr 4 Modified Loading Assembly	8887279	5
Incr 5 Modified Loading Assembly	8887280	5
Igniter Assembly	8887282/9295114	5
Flash Reducer Assembly	8887287	5
Flash Reducer Assembly	8887291	5
Base Charge & Body Assembly	8887864	5
Charge, Propelling, 155MM M3A1 for Cannon M1, M45, M126, M185, M199 (in sealed containers)	8887277/7548187	10

4.3.2 Inspections to be performed. The First Article parts submitted in accordance with paragraph 4.3.1 will be subjected by the Government to any or all the requirements of the applicable drawings and specifications and to the inspections of Table II.

4.3.3 Rejection. See MIL-A-48078.

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TABLE II - First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-C-60418B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER / NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Propelling Charge Components and Assemblies				
	Flash Reducer (Dwg 8887288) Examination for Defects	5		3.2	4.4.2.1
	Flash Reducer, Increment 1 (Dwg 8887286) Examination for Defects	5		3.2	4.4.2.2
	Flash Reducer (Dwg 8887292) Examination for Defects	5		3.2	4.4.2.3

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TABLE II - First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60418B (AR)

PARAGRAPH CATEGORY	TITLE: Propelling Charge Components and Assemblies	SHEET 2 OF 3			DRAWING NUMBER / NEXT HIGHER ASSEMBLY
	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Reducer Flash (Dwg 8887290) Examination for Defects	5		3.2	4.4.2.4
	Igniter End (Dwg 8887283) Examination for Defects	5		3.2	4.4.2.5
	Igniter Increment 1 (Dwg 8887281) Examination for Defects	5		3.2	4.4.2.6
	Base Body Charge 1 (Dwg 9204942) Examination for Defects	5		3.2	4.4.2.7
	Base Charge 1 Body & Igniter Incr Assembly (Dwg 9204943) Examination for Defects	5		3.2	4.4.2.8

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TABLE II - First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60418B (AR)

PARAGRAPH CATEGORY	TITLE / EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	DRAWING NUMBER / NEXT HIGHER ASSEMBLY / PARAGRAPH REFERENCE /INSPECTION METHOD
	Propelling Charge Components and Assemblies		SHEET 3 OF 3		
	Base Charge 1 Loading Assembly (Dwg 8887278) Examination for Defects	5		3.2	4.4.2.9
	Body Increment (Dwg 9221256) Examination for Defects	5		3.2	4.4.2.10
	Increment Loading Assembly (Dwg 9221257) Examination for Defects	5		3.2	4.4.2.11
	Increment 4 Modified Loading Assembly (Dwg 8887279) Examination for Defects	5		3.2	4.4.2.12
	Increment 5 Modified Loading Assembly (Dwg 8887280) Examination for Defects	5		3.2	4.4.2.13
	Charge Propelling, 155MM, M3A1 for Cannon, M1, M45, M126, M185, M199 (Dwg 8887277/7548187) Examination for Defects	10		3.2	4.4.2.14/ 4.4.2.15/ 4.4.2.16

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of propelling charge assemblies shall contain:

a. Propellant from not more than one lot number from one manufacturer.

b. CBI or BPI from not more than one lot number from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60418B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Flash Reducer		8887288
			NEXT HIGHER ASSEMBLY 8887287

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Cloth improper		0.65%	3.2	4.5.4
202	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.8	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60418B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8887286
4.4.2.2	Increment 1 Flash Reducer				NEXT HIGHER ASSEMBLY 9204943
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Assembly damaged to the extent that salt can escape		0.40%	3.2	Visual
102	Weight of salt		0.40%	3.2	Balance
Minor					
201	Any seam or opening incompletely stitched, including broken threads		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60418B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Flash Reducer		8887292
			NEXT HIGHER ASSEMBLY 8887291

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Cloth improper		0.65%	3.2	4.5.4
202	Marking misleading, missing or unidentifiable		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.8	Visual

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12

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60418B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Flash Reducer				8887290
					NEXT HIGHER ASSEMBLY 8887279/8887280
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Assembly damaged to the extent that salt can escape		0.40%	3.2	Visual
102	Weight of salt		0.40%	3.2	Balance
Minor					
201	Any seam or opening incompletely stitched		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60418B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Igniter, End				8887283
					NEXT HIGHER ASSEMBLY 8887282
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Cloth improper		0.65%	3.2	4.5.4
202	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60418B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Igniter, Increment 1				8887281
					NEXT HIGHER ASSEMBLY 9204943
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Assembly damaged to the extent that BPI or CBI can escape without force		100%	3.2	Visual
Major 101	Weight of CBI or BPI		100%	3.3	4.5.2
Minor 201	Cloth improper		0.65%	3.2	4.5.4
202	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60418B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Body, Base Charge 1				9204942
					NEXT HIGHER ASSEMBLY
					8887864
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Tying strap missing, insecure or improperly assembled		0.40%	3.2	Visual/Manual
Minor					
201	Cloth improper		0.65%	3.2	4.5.4
202	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60418B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Base Charge 1 Body and Igniter Increment Assembly	1 OF 1	9204943
			NEXT HIGHER ASSEMBLY: 8887278

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Igniter charge increment missing		100%	3.2	Visual
Major	None defined				
Minor	None defined				

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60418B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Base Charge 1 Loading Assembly				8887278
					NEXT HIGHER ASSEMBLY 8887277
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Assembly damaged to the extent that CBI or BPI can escape without force		100%	3.2	Visual
Major					
101	Flash reducer increment missing		0.40%	3.2	Visual
102	Assembly damaged to the extent that propellant can escape without force		0.40%	3.2	Visual
103	Propellant weight		100%	3.3	4.5.1
Minor					
201	Any seam or opening incompletely stitched, including broken threads		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60418B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9221256
4.4.2.10	Body, Increment				NEXT HIGHER ASSEMBLY 9221257
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor 201	Cloth improper		0.65%	3.2	4.5.4
202	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60418B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Increment Loading Assembly				9921257
					NEXT HIGHER ASSEMBLY 8887277/79/80
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Assembly damaged to the extent that propellant can escape without force		0.40%	3.2	Visual
102	Propellant weight		100%	3.3	4.5.1
Minor					
201	Any seam or opening incompletely stitched		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60418B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.12	Increment 4 Modified Loading Assembly		8887279
			NEXT HIGHER ASSEMBLY 8887277

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Flash reducer increment missing		0.40%	3.2	Visual
102	Assembly damaged to the extent that propellant can escape without force		0.40%	3.2	Visual/Manual
Minor					
201	Any seam or opening incompletely stitched, including broken threads		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60418B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8887280
4.4.2.13	Increment 5 Modified Loading Assembly				NEXT HIGHER ASSEMBLY 8887277
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Flash reducer increment missing		0.40%	3.2	Visual
102	Assembly damaged to extent that propellant can escape without force		0.40%	3.2	Visual/Manual
Minor					
201	Any seam or opening incompletely stitched		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60418B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.14	Charge, Propelling, 155MM, M3A1 for Cannon M1, M45, M126, M185, M199		8887277
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Assembly damaged to the extent that BPI or CBI can escape without force		100%	3.2	Visual/Manual
Major					
101	Propelling charge assembly weight		100%	3.3	4.5.5
102	Proving Ground	20		3.5	4.5.6
103	Dimensional gage failure		100%	3.4	Gage
104	Any component improperly assembled		100%	3.2	Visual
105	Assembly damaged to extent that propellant can escape without force		100%	3.2	Visual
Minor					
201	Igniter protector cap missing or improperly attatched		0.65%	3.2	Visual
202	Strap loose, broken or insecurely tied		0.65%	3.2	Visual/Manual
203	Evidence of poor workmanship		0.65%	3.8	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60418B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.15	Container, Prior to Sealing				7548187
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Any packing components missing		0.65%	3.2	Visual
202	Charge cannot be removed by hand		0.65%	3.2	Visual/Manual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60418B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.16	Container, Sealed				7548187
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Test plug missing		100%	3.2	Visual
102	Test plug assembly torque		0.40%	3.6	Test
Minor					
201	Test plug not greased (to be checked after air pressure test and prior to reassembly)		0.65%	3.2	Visual
202	Car seal missing or incorrectly applied		0.65%	3.2	Visual
203	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual
204	Cover incompletely engaged		0.65%	3.2	Visual

NOTES:

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4.4.3 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment design in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and paragraph 6.3 of this specification.

4.5 Test methods and procedures

4.5.1 Propellant weight (See 3.3). The inspections below are to be conducted by inspection personnel using equipment designated for inspection purposes and shall be clearly distinguished from production operations. Any increment which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Inspection equipment shall be in accordance with 4.4.3.

4.5.1.1 Method 1. The loaded increments will be inspected 100 percent. The weight of the propellant will be determined by subtracting an average weight of the empty increment bag or the actual weight as marked on the individual increment bag from the loaded increment weight.

4.5.1.2 Method 2. The weight of the propellant will be inspected 100 percent prior to loading the increment on a different balance. In addition, the propellant weight of the loaded increment will be inspected using a sampling plan from MIL-STD-1235 with an AQL of 0.40 percent. The weight of the propellant will be determined by subtracting an average weight of the empty increment bag or the actual weight as marked on the individual increment bag from the loaded increment weight.

4.5.2 Weight of CBI or BPI in base igniter assembly. The inspections below are to be conducted by inspection personnel using equipment designated for inspection purposes and shall be clearly distinguished from production operations. Any assembly which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Inspection equipment shall be in accordance with 4.4.3.

4.5.2.1 Method 1. The loaded assemblies will be inspected 100 percent. The weight of the CBI or BPI will be determined by subtracting an average weight of the empty assembly or the actual weight as marked on the individual assembly from the loaded assembly weight.

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4.5.2.2 Method 2. The weight of the CBI or BPI will be inspected 100 percent prior to loading the assembly on a different balance. In addition, the CBI or BPI weight of the loaded assembly will be inspected using a sampling plan from MIL-STD-105 or MIL-STD-1235 with an AQL of 0.40 percent. The weight of the CBI or BPI will be determined by subtracting an average weight of the empty assembly or the actual weight as marked on the individual assembly from the loaded assembly weight.

4.5.3 Air pressure test. Each container shall be tested as specified on the appropriate drawing. A pressure drop equivalent to or greater than the applicable requirement shall be cause for rejection and removal from the lot.

4.5.4 Cloth. At the time cloth is introduced to the sewing and/or cutting operation, an identification will be made for each roll to verify that proper material is used. Any cloth failing to be identified as proper material in accordance with applicable drawings and specifications shall be rejected from the lot.

4.5.5 Charge, propelling assembly weight. The average weights of the empty increment assembly and nominal weights of CBI or BPI and Potassium Nitrate/Sulfate together with the nominal propellant weights as specified in the loading authorization shall be subtracted from the assembled propelling charge assembly weight. If the resulting weight deviation is greater than the requirement, the propelling charge will be classified defective and removed from the lot.

4.5.6 Proving ground (See 3.5). This test shall be conducted at a Government Proving Ground in accordance with the applicable acceptance test procedure using M107 projectiles inert loaded within 0.1 pound of an as fired weight of 95.0 pounds using Primer, MK2A4 and fired in cannon M1. Prior to firing, all charges shall be temperature conditioned in a large conditioning box with adequate air circulation at the required temperature ($\pm 2^{o}F$) for a minimum of 24 hours and assurance will be made that, when fired, the charges are at the conditioning temperature. Copper crusher gages (2) shall be used to measure the maximum chamber pressure of each charge. Velocity coils will be used to measure the muzzle velocity of each round.

4.5.6.1 Muzzle velocity and muzzle velocity standard deviation. Determination of acceptance will be in accordance with MIL-STD-414, Section B, Table B3, AQL of 0.65%. All muzzle velocity and muzzle velocity standard deviation samples will be ballistic

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tested at 70°F. Ten (10) zone 1 and ten (10) zone 5 charges will be fired for muzzle velocity and muzzle velocity standard deviation. If the sample is determined unacceptable the lot will be rejected. In addition, if the velocity standard deviation of the sample multiplied by the factor 0.72 exceeds the applicable lot requirement, the lot shall be rejected. Ten (10) calibration charges (five at zone 1 and five at zone 5) will be fired intermixed with the test samples and the muzzle velocity and pressure of all test charges will be corrected to standard conditions.

4.5.6.2 Pressure. If any propelling charge individual maximum pressure, when corrected to standard conditions, exceeds the applicable requirement, the lot shall be rejected.

5. PACKAGING

5.1 Packing and marking. The propelling charges shall be packed, sealed, tested and marked in accordance with the specified requirements of drawing 7548187.

6. NOTES

6.1 Intended use. The propelling charges covered by this specification are intended for use in the 155MM Cannon M1, M45, M126, M185 and M199.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of designs for approval. See 6.2.3 of MIL-A-48078. Submit equipment designs, as required, to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

6.4 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic may be measured or gaged to determine acceptability.

6.5 Inspection lot size. It is noted that the size of inspection lots of components, assemblies, or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

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6.6 Proving ground. A copy of all proving ground data should be submitted to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-R and DRDAR-LCA-G, Dover, New Jersey 07801 and to Master Repository, US Army Armament Materiel Readiness Command, ATTN: DRSAR-QAD, Rock Island, IL 61299.

6.7 Charge, propelling assembly weight. This requirement applies to the load plant at the time of loading only. Due to subsequent change in tare weight of the components because of variation in material and moisture content, this test cannot be repeated after initial manufacture to the degree of accuracy required.

6.8 Test plug assembly torque. This requirement applies to the load plant at the time of loading only. Due to creep of the mechanical properties of the test plug, this test cannot be repeated after initial manufacture.

6.9 Method of inspection submission. Methods of inspection for cloth for ignition components shall be submitted to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-R, Dover, New Jersey 07801 for approval.

6.10 Data cards. A copy of all data cards should be submitted to Commander, ARRADCOM, ATTN: DRDAR-QAR-R and DRDAR-LCA-G, Dover, New Jersey 07801 and to Master Repository, US Army Armament Materiel Readiness Command, ATTN: DRSAR-QAD, Rock Island, IL 61299.

6.11 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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MIL-C-60418B Charge, Propelling, 155MM. M3A1 Loading, Assembling and Packing

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